

Discovering Patterns in Palmer Penguins

STAT 109 · Project 3 sample one-page summary

Sample student report (illustrative)

June 1, 2026

Introduction

How to penguin characteristics differ between species? Using data from one Antarctic sample (available at TidyTuesday week [2020-07-28](#)), I analyzed two traits between three species at several locations. The full dataset has 344 penguins and 8 columns. I focused on species (categorical), island (categorical), flipper length in mm (numerical), and body mass in g (numerical). Two rows were missing flipper length and body mass, so after a missingness check I dropped them and analyzed the 342 penguins with complete data on all four variables

Descriptive Analysis

Summary	Value
Proportion of Gentoo Penguins	124/342 = 0.36 (36%)
Proportion on Biscoe island	168/342 = 0.49 (49%)
Mean flipper length (SD)	200.9 (SD)
Mean body mass (SD)	4202 (802)
Pearson r (flipper length vs body mass)	0.87

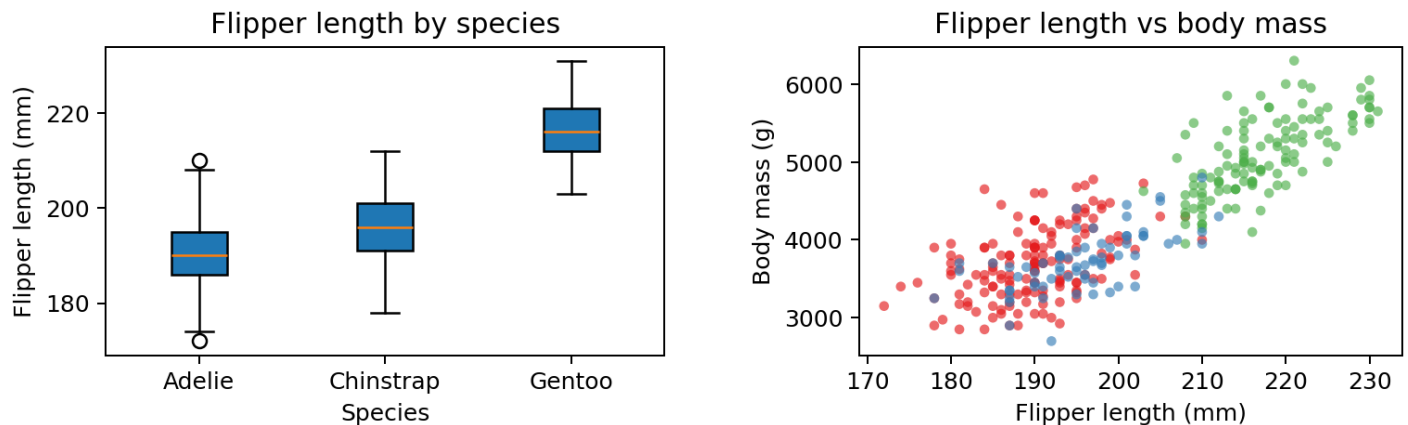


Figure 1: Left: flipper length by species. Right: flipper length vs body mass (points colored by species in the notebook).

Conclusions

1. Gentoo penguins have an average flipper length of about 217 mm (SD 6), which is longer than the other two species. Adelie and Chinstrap with Adelies having a mean length of about 190 mm (SD 7), and Chinstrap a mean length of about 196 mm (SD 7).
2. Flipper length and body mass have a strong, positive, linear relationship in this sample ($r = 0.87$): heavier penguins in this sample tend to have longer flippers.
3. Gentoo make up about 36% of the sample, and the row proportions in my two-way table show essentially all of them live on Biscoe island.
4. Penguins are heavier than I was expecting: the mean body mass in my sample was 4202 grams, which is about 9 pounds, similar to a light bowling ball.